Saksham Sahu

+91-7906552119 | sakshamsahu77783@gmail.com | linkedin.com/in/sahusaksham10 | github.com/Saksham0121
Portfolio: sakshamsahu.me | Faridabad, India

EDUCATION

Bennett University Aug 2023 – Jun 2027
Bachelor of Technology in Computer Science Engineering; CGPA: 8.72 Noida, India

EXPERIENCE

DRDO, Ministry of Defence May 2026 - Jul 2026
Generative AI Intern

- Engineered secure, locally hosted **Advanced RAG** system on a **Unix/Linux** environment using LLMs, vector databases and embedding models, achieving **92.5% retrieval accuracy** on a **460-question** benchmark over a confidential knowledge base
- Developed **Hybrid RAG** architectures with semantic search, keyword retrieval and reranking while reducing query latency by **~25%**
- Built scalable document processing pipelines handling **11,000+ pages**, including chunking, embedding generation, indexing, and structured **logging** for debugging, reducing hallucinations through multi-stage retrieval

Imarticus Learning Jun 2025 - Jul 2025
Data Analyst Intern

- Led a team of **5 analysts** to develop **interactive dashboards** delivering key business insights to stakeholders
- Analyzed **50,000+ row** business datasets using **MySQL, Python and Power BI** to extract actionable insights
- Visualized insights using **Matplotlib and Seaborn** for weekly stakeholder reports

SKILLS

Languages : C++, Python, Java, JavaScript, TypeScript
Web & Backend : React.js, Spring Boot, Spring Data JPA, Next.js, Node.js, Hibernate, Express.js, REST API
Databases : MySQL, MongoDB, PostgreSQL, Vector Databases
Tools : GitHub, Postman, Power BI, Figma
Cloud & DevOps : AWS(ECR, ECS, EC2, ALB), Docker, Git, CI/CD pipelines
Gen AI : RAG, LLMs, LangChain, LangGraph, ChromaDB
Core CS : Data Structures & Algorithms, OOPs, DBMS, Operating System, Computer Networks, System Design

PROJECTS

Parakh Trading Intelligence | Java 21, Spring Boot 3, Kafka, PostgreSQL, Redis, k6, OpenTelemetry Sep 2026

- Developed an event-driven trading platform in **Java 21** & **Spring Boot 3**, using **Apache Kafka** and Virtual Threads to ingest live market feeds, run asynchronous tasks and trigger real-time trade alerts
- Built a historical **backtesting** simulation engine and **TimescaleDB** pipeline to test trading strategies across multi-year candle data, calculating PnL, win rates and technical indicators (RSI, MACD, EMA)
- Reduced p95 API latency by **40%** (984ms to 587ms) and p99 by **61%** (2.54s to 996ms) at **120 concurrent VUs** in **k6** by optimizing **Redis** caching and database connection pooling; traced using **OpenTelemetry**

Askra AI | React.js, FastAPI, FAISS, BM25, MongoDB, JWT, Docker, Grafana k6 Jun 2026

- Architected an enterprise **Agentic RAG** platform for private-document Q&A with a **7-layer pipeline** and LLM-as-judge self-correction, raising answer accuracy from **68%** to **92%** over a baseline RAG on a ground-truth QA benchmark (LLM-judged)
- Engineered hybrid **FAISS + BM25** retrieval with cross-encoder reranking, running both searches concurrently via thread futures for **354ms** P95 TTFT; integrated a local OCR model to auto-process scanned PDFs
- Shipped async **FastAPI** + **MongoDB** backend with **JWT** auth, department-level **RBAC** and **SSE** streaming from Groq LLMs (∼**100ms TTFT**); cut cold-start latency **83%** via startup model pre-warming; containerized with **Docker** and **Nginx**

Social-ish | React.js, Node.js, MongoDB, WebSocket, Gemini API Apr 2025

- Built and deployed a **full stack social platform** for introverts with interest based matching and real time WebSocket chat
- Designed a scalable REST API with **JWT authentication** and MongoDB indexing for fast interest matching queries

ACHIEVEMENTS

Leetcode/amSaksham Ongoing
*Solved **700+** DSA problems*

Innovate 2.0: Hack to Build – JIIT Noida 2024
National level Winner among 200+ teams

TechArena 2025 Project Showcase 2025
Secured 3rd position